

Introduction

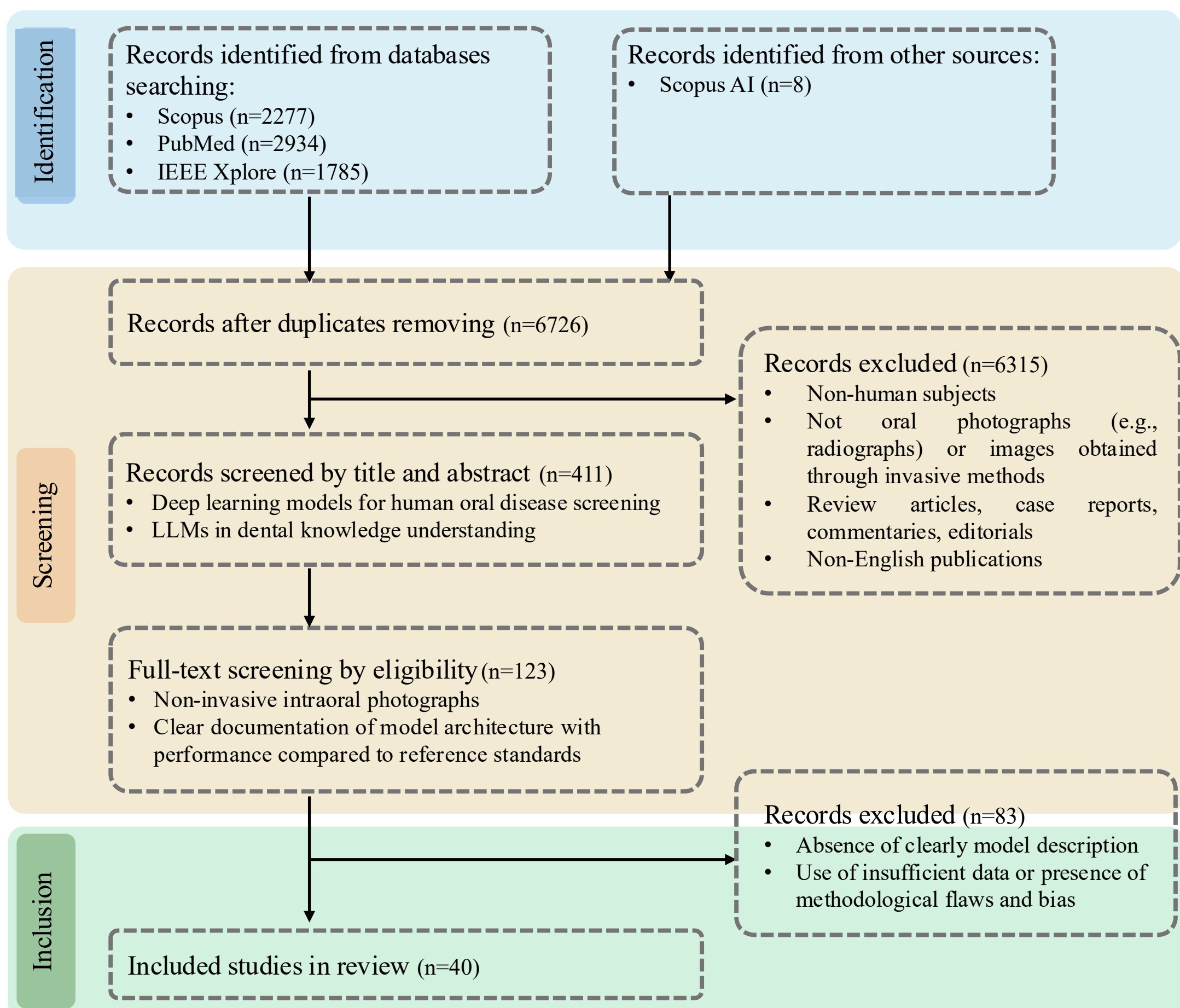
In response to the growing need for early detection, preventive care, and personalized treatment, AI models based on intraoral photographs may provide accurate, consistent, and efficient screening tool for dental diseases. The advent of generative AI, particularly large language models (LLMs), holds significant promise for enabling personalized oral health surveillance. Evaluating the screening performance of existing AI models and the domain knowledge capabilities of LLMs is essential for advancing precision dentistry.

Objectives

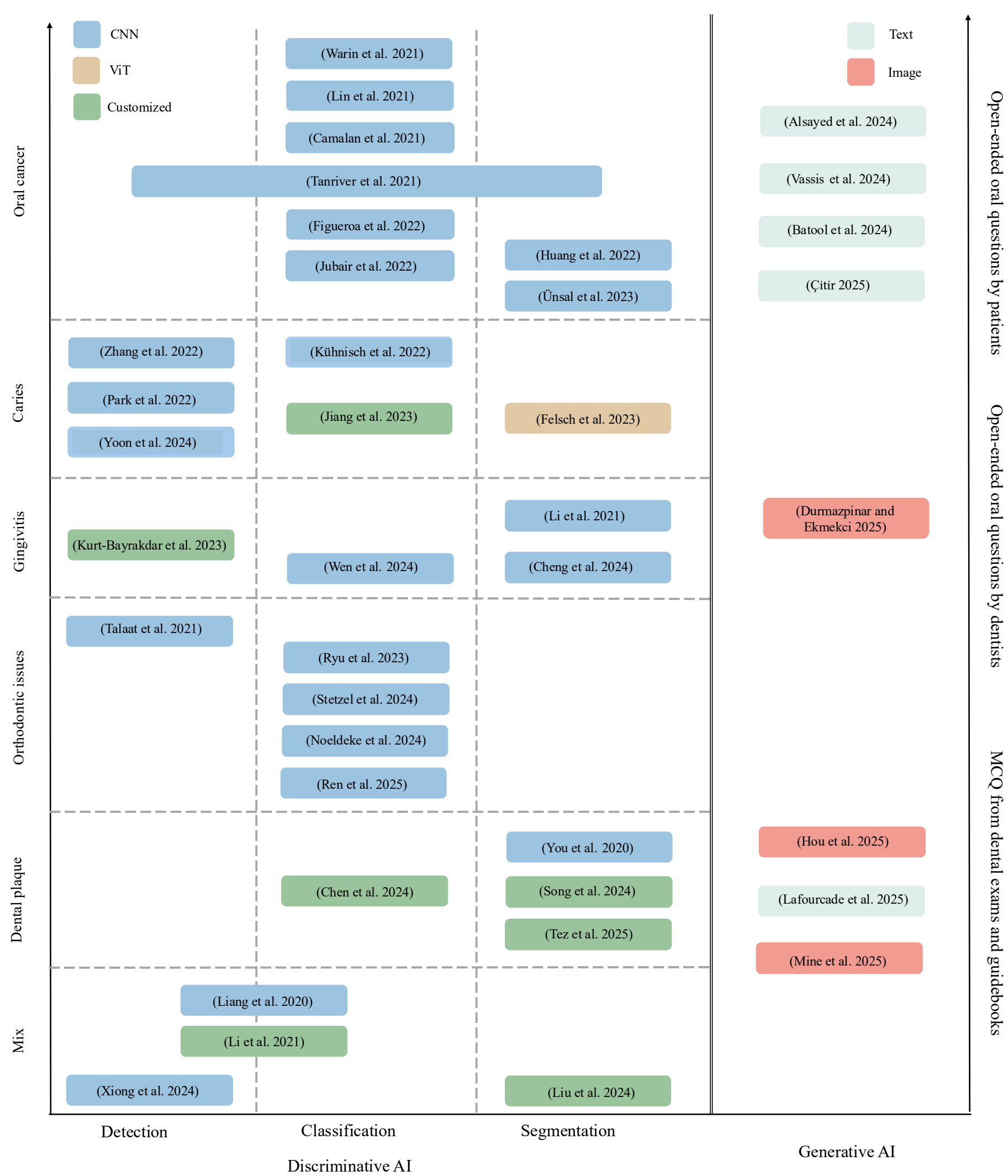
- Review AI performance in oral disease screening based on intraoral photographs;
- Review performance of LLM in dental consultation;
- Explore better modelling pipeline from data processing to model design and performance evaluation, and dental domain LLM refinement strategy.

Studies Overview

Study Screening & Selection



Study Characteristics



- **Oral disease screening:** 32 studies use **discriminative AI models** for classification, detection, or segmentation of dental caries, gingivitis, oral cancer, dental plaque, malocclusion traits, and various oral conditions.
- **Dental consultation:** 8 studies evaluate existing **generative AI (LLMs)** on dental knowledge understanding via open-ended or multiple-choice questions, with formats spanning text and image modalities.

Study Quality Assessment

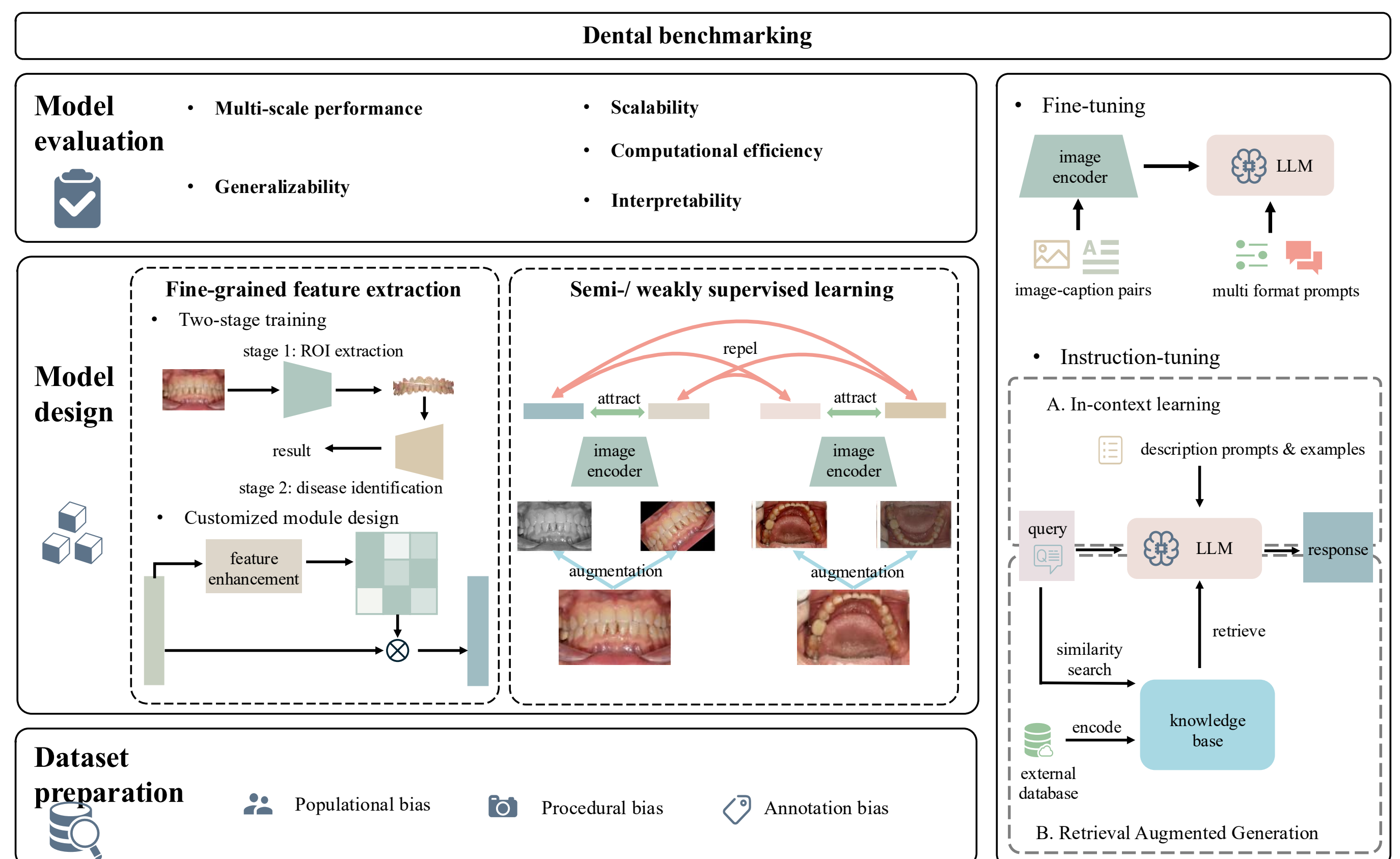
Section	Subsection	No	Item	Section	Subsection	No	Item
Title		1	Relevant topic	Data		27	Data characteristics
Abstract		2	Structured summary			28	Model selection
		3	Scientific background			29	Ablation study
Introduction		4	AI approach rationale			30	Robustness analysis: detailed performance
		5	Study objective			31	Robustness analysis: uncertainty
	Study design	6	Adherence to guidelines			32	Robustness analysis: failure analysis
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		8	Study nature			34	Testing with external data
		9	Eligibility criteria			35	Testing with dentists
		10	Method pipeline			36	Overall findings
		11	Data source			37	Limitations
		12	Data overlap			38	Photography quality
		13	Sample size			39	Comparison with previous works
		14	Data split			40	Practical implications
		15	Data preprocessing			41	Dataset open-sourced
		16	Data imbalance			42	Code open-sourced
		17	Reference standard			43	Model open-sourced
		18	Annotation definition			44	Application open-sourced
		19	Annotation tool				
		20	Annotator detail				
		21	Annotator consistency				
		22	Algorithm details: model				
		23	Algorithm details: software				
		24	Training and tuning details: parameter initialization				
		25	Training and tuning details: training approach				
		26	Performance metrics				

A merged and modified medical imaging AI checklist^[1,2] was applied to guide risk of bias assessment on studies regarding intraoral photo-based AI models.

Key Findings

- Significant variability in data preprocessing, annotation, models, and evaluation;
- AI achieves dentist-level performance in caries and plaque identification;
- Limited accuracy in assessing disease severity and fine-grained malocclusions, especially without 3D data;
- LLMs demonstrate strong general dental knowledge, but lack visual reasoning, complex decision-making, and reliability due to 'hallucinations'.

Discussion



- **Dataset preparation** for intraoral photographs
- **Model design** for fine-grained feature extraction combined with semi- or weakly supervised learning
- **Comprehensive evaluation framework**
- **Dental-specific multimodal LLM design**

Conclusion

This study highlights the potential of intraoral image-based AI for disease screening and dental-specific LLMs for multimodal understanding, underscoring the future of AI-driven oral health surveillance.

Refs: [1] Mongan J, Moy L, Kahn CE *et al.* 2020. Checklist for artificial intelligence in medical imaging (CLAIM): A guide for authors and reviewers. *Radiol Artif Intell.* 2(2):e200029.
[2] Kocak B, Baessler B, Bakas S, *et al.* 2023. Checklist for evaluation of radiomics research (CLEAR): A step-by-step reporting guideline for authors and reviewers endorsed by ESR and EUSOMI. *Insights Imaging.* 14(1):75.